

Recipe Determination using ML

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Current Process

Client(FG) manufacture FeSi Alloys with custom constituent elements in prescribed proportions by weight to meet their varied customer requirements.

In order to achieve this, FG uses its existing processes to get the base molten metal/alloy and add various additives to induce the required constituent elements into the molten metal to achieve the desired final composition.

In addition, Additives when added to the molten metal can result in exothermic/endothermic reactions resulting in increase/decrease of temperature of the molten metal/alloy. FG also needs to ensure the temperature of the molten metal at optimal levels before solidification after adding additives.

Also, Additives will not add 100% of the Constituent elements to the final product and it is given only a certain amount of the constituent element will be recovered from a given additive. This is called Recovery Percentage. These percentages also play an important role in arriving at the final weights of the additives that are required to be added.

Current Process

At present, FG has an application which uses an iterative approach (Trial & Error) to determine the weights of various additives that need to be added to the base molten metal to achieve the required final composition and also taking into account the temperature effect and recovery that each additive has.

Problem Statement: While the current approach is working, the application has aged both in terms of infrastructure and technology. Also depending on the number of constituent elements, the number of iterations required change to arrive at the final composition.

Given the critical nature of the application, it is decided to modernise the stack so the application can become reliable and scalable across multiple plants.

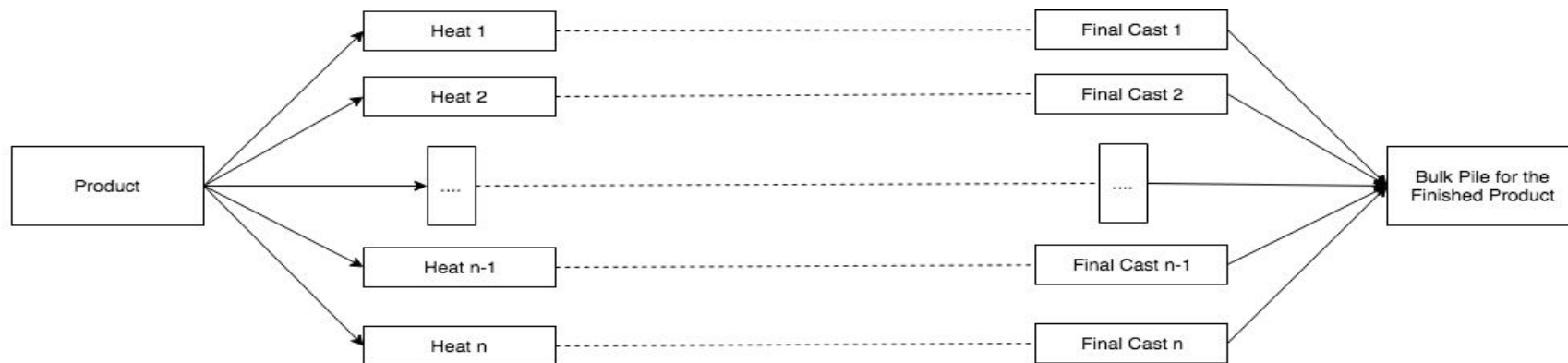
As part of the modernisation, It is being intended/explored to use technology (Machine learning) to replace the existing iterative approach to arrive at the weights of the various additives.

Objective

Explore the feasibility of developing a ML Algorithm based on the available data to determine the weights of the additives to manufacture the final product with constituent elements in required proportions while taking into account the recovery percentages and the temperature effect of each additive to maintain the optimal temperature of the final product prior solidification.

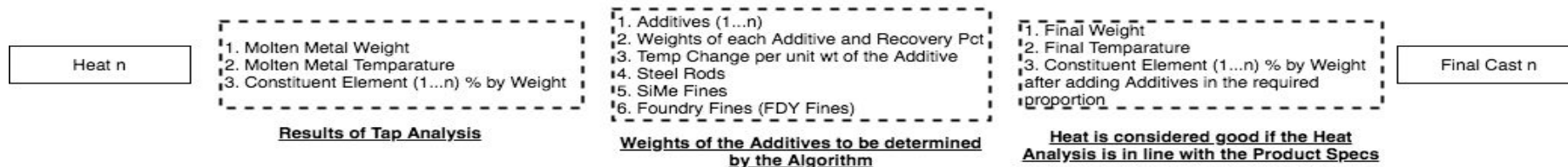
Though there are empirical/theoretical recovery percentages available from Additive Analysis, existing system capture actual recoveries of the constituent elements from additives and use them for calculations in the iterative approach.

Schematic Diagram of the approach



All heats are considered as good heats

For this schematic diagram



Representation Equation

$$FP (FW, e_1, e_2, \dots, e_n, t_{fp}) = \sum_{n=1}^n Wt (Additive_n * Recovery Percentage_n) + \sum_{n=1}^n Wt (Additive_n) * tc_n + BP (BW, e_1, e_2, \dots, e_n, t_{bp})$$

FP - Final Product

BP - Base Product (Molten Metal from the Furnace)

FW - Final Weight

BW - Molten Metal Weight

e - % of Constituent Elements by weight basis such as Al, Mg, Ba, Ln etc.,

Additive - Additive Materials such as Mg Ingots, Al Ingots, Barium Silicide, SiMe Fines etc.,

Recovery Percentage - Percentage of element that can be recovered from the Additives

t_{fp} - Target temperature of the final product

t_{bp} - Molten Metal Temperature

t_{cn} - Temperature Change per unit weight of the Additive

ML Approach - Understanding Existing Data and its Sufficiency

Table/File Name	# of Records	Date Range Available
HEATDETAILTABLE	90228	01/01/2018 - 09/28/2023
HEATDETAILTABLE_TAPANAL	90228	01/01/2018 - 09/28/2023
HEATDETAILTABLE_HTANAL	90228	01/01/2018 - 09/28/2023
HEATDETAILTABLE_RAWMATERIALS	1305897	01/01/2018 - 09/28/2023
HEATDETAILTABLE_RAWMATERIALS_CTD	1305897	01/01/2018 - 09/28/2023
HEATDETAILTABLE_PREADDHTREC	90228	01/01/2018 - 09/28/2023
MATERIALFILE	3000	06/28/2023 - 10/14/2023
MATERIALFILE_WEIGHTS	3000	06/28/2023 - 10/14/2023
PRODUCTS_HDR	1676	Not Applicable (Product Master)
PRODUCTS_CHEM_A	1676	Not Applicable (Product Master)
LADLEADDDETAILFILE	216	Not Applicable (Additive Master)
LADLEADDDETAILFILE_RECOVERY	216	Not Applicable (Additive Master)